Newborough Warren

Interactive Web Tools

User Manual

Forecaster · Scenario Viewer · Seasonal Extremes Scatter

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Contents

Introduction

This manual provides step-by-step guidance for operating the three interactive web tools produced by the Newborough Warren hydrology research project (Hollingham 2026). The tools are designed to be opened in any modern web browser with no installation or server infrastructure required.

- **Part A covers the Groundwater Flooding Forecaster,** which provides per-well flood-risk assessments based on observed water-table depths and a rainfall multiplier.
- **Part B covers the Hydrological Scenario Viewer,** an interactive map tool for exploring the effects of climate change and forest management scenarios on groundwater levels across the site.
- **Part C covers the Seasonal Extremes Scatter,** a diagnostic chart that plots each well's mean seasonal water-table extremes against published eco-hydrological thresholds. For details on the underlying model equations and data architecture, see the companion Technical Note.

Part A

Groundwater Flooding Forecaster

A1. Introduction

The Groundwater Flooding Forecaster is a self-contained, single-file HTML tool that provides per-well flood-risk assessments for the Newborough Warren dipwell network. It is designed to be opened in any modern web browser, on desktop or mobile, with no installation required.

The forecaster answers three linked questions for any selected dipwell:

1. What winter peak water level is expected, given the current summer minimum and a chosen rainfall scenario?
2. What summer minimum follows from that winter peak?
3. How much cumulative rainfall is needed to bring groundwater to the surface (P_flood)?

A2. Opening the Forecaster

Open forecaster.html in a web browser. The page loads instantly because all well data, map geometry, and model coefficients are bundled into the file. No server or internet connection is required for the core functionality.

On first load the tool will attempt to fetch live rainfall data from the Met Office (see Section A5). This requires an internet connection but is optional; the tool falls back to climatological defaults if the fetch fails.

A3. Interface Layout

The interface is arranged in three panels beneath a header bar and a live-data banner:

Panel	Purpose
Left sidebar	Well selection (dropdown grouped by cluster), observed-depth input, and well metadata (coordinates, elevation, cluster assignment).
Centre map	SVG plan of Newborough Warren showing every dipwell as a colour-coded dot. A hillshade base layer and KML feature overlays provide spatial context. A rainfall-multiplier slider and map legend are overlaid.
Right panel	Three forecast cards for the selected well, plus a method note. Tabs at the top switch between Cluster average and Per-well SSM sources.

On narrow screens (< 960 px) the three panels stack vertically.

A4. Selecting a Well

Wells can be selected in two ways:

- Dropdown menu (left sidebar): wells are grouped by cluster (C1–C5) with descriptive labels. Wells marked with an asterisk (*) are nearest-type assignments, meaning they sit outside the SSM operational domain and forecasts should be interpreted with caution.
- Map click: click any dot on the centre map to select that well. The selected well is highlighted with a larger radius.

When a well is selected the observed-depth field pre-populates with the cluster long-term summer minimum. Edit this value to match your current field observation.

A5. Live Met Office Data

On load, the forecaster fetches the Met Office Historic Station Data file for RAF Valley (valleydata.txt). If the fetch succeeds, a banner below the header shows:

- Cumulative winter rainfall (Oct–Mar) for the current and/or most recently completed hydrological winter.
- The observed rainfall expressed as λ (lambda), the ratio of observed to climatological winter total.

Two preset links appear above the slider allowing you to set the rainfall multiplier to the live λ or to the climatological default (λ = 1.00).

A5.1 Manual fallback

If the live fetch fails (CORS restriction, no internet, server down), a Retry button and an Enter rainfall manually button appear. The manual option opens a modal where you can paste the full contents of valleydata.txt copied from a browser tab. The parser extracts the same monthly totals and populates the banner identically.

A6. The Rainfall Multiplier (λ)

The slider at the bottom-right of the map controls the assumed winter rainfall multiplier λ, ranging from 0.60× (dry winter) to 2.00× (exceptional winter). As you drag it:

- The map dots re-colour in real time to reflect updated P_flood vulnerability.
- The forecast panel recalculates all three forecasts.
- A descriptive label updates: Dry winter, Climatological, Wet winter, Very wet winter, or Exceptional winter.

A7. Understanding the Forecasts

The right panel displays three forecast cards, each with a numeric result, a coloured badge, and supporting context:

A7.1 Forecast 1 — Next winter peak

Predicts the shallowest winter water-table depth given the entered summer observation and the assumed rainfall. The badge indicates whether the SD15b favourable-condition threshold (≤0.10 m below ground) would be met:

- SD15b met (green): peak reaches within 0.10 m of the surface.
- SD16 only (amber): peak reaches 0.10–0.25 m below ground.
- Below SD16 (red): peak remains deeper than 0.25 m.

A7.2 Forecast 2 — Following summer minimum

Projects the deepest summer water-table depth from the cluster-mean winter peak and climatological summer rainfall (Apr–Sep). The badge uses the SD15b viability threshold:

- SD15b viable (green): minimum ≤0.61 m below ground.
- SD16 only (amber): 0.61–0.98 m.
- Below SD16 (red): deeper than 0.98 m.

A7.3 Forecast 3 — P_flood to surface

Calculates the cumulative rainfall (in mm) over the cluster's horizon period needed to bring groundwater from its current depth to the surface (0 m). The result is also expressed as λ, the multiple of climatological rainfall:

- Reachable (green): λ < 1.0 — surface flooding possible under normal rainfall.
- Wet winter required (amber): λ 1.0–1.3.
- Exceptional winter (red): λ 1.3–2.0.
- Structurally unreachable (grey): λ > 2.0 — flooding essentially impossible.

A8. Cluster Average vs Per-Well SSM

When a well has its own SSM coefficients (most do), a pair of tabs appears above the forecast cards:

- Cluster average: uses the block-level seasonal transfer function (Tables 8 & 9 in Hollingham 2026) for Forecasts 1 and 2, and the cluster-centroid P_flood equation for Forecast 3.
- Per-well SSM: uses each well's own lag-1 state-space model coefficients to iterate the monthly recurrence forward.

Whichever tab is active, the other source's estimate appears as a comparison row beneath the main forecast.

A9. Reading the Map

Each well dot is coloured according to the P_flood ratio to the SD15b threshold (−0.10 m), scaled by the current λ slider setting:

Colour	Category	λ range
Green	Reachable	< 1.0× climatology
Amber	Wet winter required	1.0–1.3×
Red	Exceptional winter	1.3–2.0×
Grey	Structurally unreachable	> 2.0×

A10. Nearest-Type Assignments

Some wells are flagged with an asterisk (*) and display a nearest-type assignment notice. These wells sit outside the SSM operational domain and were assigned to the nearest cluster as a best-available estimate. Forecasts for these wells should be treated as indicative rather than precise.

A11. Practical Tips

- Enter your actual field reading in the observed-depth box immediately after selecting a well; the default is a long-term average, not a live measurement.
- Use the live λ preset when available — it incorporates the actual current-season rainfall.
- Compare both the Cluster-average and Per-well SSM tabs; large divergence may indicate the well behaves atypically within its cluster.
- The forecaster works entirely offline after the initial page load; you can save the HTML file to a USB stick or phone for field use.

Part B

Hydrological Scenario Viewer

B1. Introduction

The Hydrological Scenario Viewer is an interactive HTML tool that allows you to explore how different climate and management scenarios affect groundwater levels across the Newborough Warren dipwell network. It computes per-well changes in equilibrium head (Δh) using the site's fitted state-space model coefficients and displays the results as an IDW-interpolated surface map, a cluster-level bar chart, a numeric data table, and summary metric cards.

The viewer is generated by Script 19 (19_spatial_groundwater.py) and contains all data embedded in the file. No server or internet connection is required.

B2. Interface Layout

Panel	Purpose
Left sidebar	Scenario presets, climate sliders (winter/summer P and PET multipliers), forestry controls (interception, β_2 scaling), display toggles, and Sy mode selector. A draggable splitter adjusts its width.
Season tabs	Toggle between Annual, Winter (Nov–Mar), and Summer (May–Sep) views. All forecasts, map, and table update accordingly.
Map	Canvas-rendered IDW interpolation surface of Δh (or absolute head, or depth-below-surface). Hillshade basemap with KML overlays.
Bar chart	Per-cluster Δh shown as horizontal bars, coloured red (drying) or blue (wetting).
Data table	Numeric breakdown per cluster: Δh, scenario head, storage shift, effective precipitation, and PET draw.
Metric cards	Summary cards for all clusters, each C1–C5, and a Forest (C4+C5) aggregate. Each shows Δh, Sy, and storage shift.

On narrow screens (< 640 px) the layout stacks vertically.

B3. Scenario Presets

Six preset buttons configure all sliders to physically meaningful combinations:

Preset	Description
Baseline	Current observed conditions. All multipliers at 1.00, forest interception at 24% (Corsican pine; Freeman 2008).
UKCP18 2050s	Central estimate (50th percentile, RCP8.5) for Wales. Wetter winters (+10% P), drier summers (−15% P), higher summer PET (+20%).
UKCP18 2080s	End-century projection. Winter P +20%, summer P −30%, summer PET +35%.
Clearfell	Complete canopy removal. Interception set to 0%, β_2 increased by ~11% (BACI-corrected Edge-tier ratio).
Broadleaf	Restocking with deciduous species. Interception reduced to 15% (Komatsu et al. 2011).
Thinning	50% canopy density reduction. Interception halved to 12%, β_2 increased by ~5% (half the BACI-corrected clearfell perturbation).

You can also adjust any slider individually after selecting a preset; the display updates in real time.

B4. Climate Sliders

Four sliders control the climate forcing multipliers independently:

- Winter P (×): scales November–March precipitation. Range 0.6–1.5.
- Summer P (×): scales May–September precipitation. Range 0.5–1.3.
- Winter PET (×): scales winter potential evapotranspiration. Range 0.8–1.3.
- Summer PET (×): scales summer PET. Range 0.8–1.5.

A warning banner appears if slider combinations exceed UKCP18 plausible ranges or create physically contradictory configurations.

B5. Forestry Controls

Two additional sliders per forest cluster (C4 and C5) control:

- Interception fraction: the proportion of rainfall intercepted by the canopy. Corsican pine baseline is 0.24; broadleaf is 0.15; clearfell is 0.00.
- β_2 scaling: multiplier on the evapotranspiration coefficient. Values > 1.0 represent enhanced atmospheric draw after canopy removal. The clearfell preset uses a BACI-corrected Edge-tier ratio of ×1.108; thinning uses ×1.054 (half-perturbation).

These controls only affect C4 (Main Forest) and C5 (Coastal Forest) wells.

B6. Map Rendering Modes

Three tabs above the map switch the colour surface:

- Δh mode (default): change from baseline equilibrium head. Blue = wetter, red = drier.
- Absolute mode: scenario-adjusted head in metres AOD, sequential colour scale.
- Depth mode: depth below ground surface (DEM minus IDW-interpolated head), anchored to Curreli et al. (2013) ecological thresholds: SD15b at 0.61 m and SD16 at 0.98 m.

In depth mode, a ridge-mask toggle suppresses interpolation over dune ridges where the DEM exceeds the local well-derived surface by more than 1.0 m.

B7. Display Toggles and Tooltips

- KML overlays: shows/hides the forest plantation, broadleaf restock area, clearfell compartment, and Llyn Rhos-ddu.
- Well labels: toggles well identifiers on the map.
- Ridge mask: in depth mode, suppresses interpolation over dune ridges.

Hovering over any well dot shows a floating tooltip with the well name, cluster, baseline head, Δh, scenario-adjusted head, specific yield, and storage shift.

B8. Practical Tips

- Start with a preset to see a scenario's overall effect, then fine-tune individual sliders.
- Use the season tabs to compare winter wetting against summer drying under the same scenario.
- Switch to Depth mode with KML overlays on to see which areas breach the SD15b/SD16 thresholds.
- The sidebar is resizable via the splitter; the map canvas is also resizable (drag the bottom-right corner).
- The viewer works entirely offline once loaded.

Part C

Seasonal Extremes Scatter

C1. Introduction

The Seasonal Extremes Scatter is an interactive chart plotting each well's mean annual summer minimum water-table depth against its mean annual winter maximum depth for the monitoring period 2005–2026. Wells are coloured by hydrogeological cluster, and Curreli et al. (2013) eco-hydrological threshold lines are overlaid. The chart provides an at-a-glance summary of which wells and clusters meet the SD15b and SD16 habitat condition targets.

The page is generated by Script 14 (14_climate_projections.py) and uses Chart.js for rendering.

C2. Opening the Page

Open 14_seasonal_extremes_scatter.html in any modern web browser. Chart.js is loaded from a CDN, so an internet connection is needed on first load (or the library can be cached). The well data is embedded inline.

C3. Reading the Chart

Each dot represents one well. The axes are:

- X-axis: Mean annual summer minimum depth (m below pipe top). More negative = deeper water table in summer.
- Y-axis: Mean annual winter maximum depth (m below pipe top). More negative = deeper water table in winter.

Wells in the upper-right quadrant have shallow water tables in both seasons (favourable for wet slack habitat). Wells in the lower-left have deep water tables year-round.

C4. Threshold Lines

Four dashed lines mark the Curreli et al. (2013) eco-hydrological thresholds:

Line	Threshold	Meaning
Vertical green	SD15b summer (−0.61 m)	Summer minimum must be shallower than 0.61 m for wet slack viability.
Vertical red	SD16 summer (−0.98 m)	Summer minimum must be shallower than 0.98 m for dry slack viability.
Horizontal green	SD15b winter (−0.10 m)	Winter maximum must reach within 0.10 m of the surface for wet slack flooding.
Horizontal red	SD16 winter (−0.25 m)	Winter maximum must reach within 0.25 m for dry slack flooding.

A well meeting both the summer and winter SD15b thresholds (upper-right of both green lines) has full wet slack habitat viability.

C5. Cluster Legend

The legend shows the colour for each cluster: C1 (Lake Edge), C2 (Dune), C3 (Western Residual), C4 (Main Forest), and C5 (Coastal Forest). Wells not assigned to a core cluster appear as “UNKNOWN” in grey.

C6. Search and Highlight

A search box above the chart lets you type a well name (e.g. “ceh36” or “nw10”). Matching wells are highlighted in orange with an enlarged dot; all other wells fade. The result text shows the well’s exact values and cluster.

Click Clear to reset the view.

C7. Practical Tips

- Use the search to locate specific wells of management interest.
- Wells plotting far to the left within C4 and C5 indicate forest-influenced water-table suppression.
- Compare the cluster distribution relative to the threshold lines to assess which parts of the site meet conservation targets.